

Touch Control

(TOUCH CONTROLLER)

The "Touch Controller" generates and stores a control voltage that is proportional to the strength of the keystroke. The control voltage available at the output of an analog memory offers a wide range of accentuation possibilities via the synthesizer's keyboard. This requires an available switching contact on the keyboard.

The primary function of a synthesizer keyboard is arguably to determine the pitch of the oscillators (VCOs). However, the output (control) voltage (CV) can also be used to track the cutoff frequency of a VCF or to modulate the frequency of VC-LFOs (see Chapter 5).

Furthermore, the keyboard and its associated electronics generate a gate pulse—(usually¹) upon every key press—that primarily serves to trigger the envelope generators.

VCAs—also "main modules" in the synthesizer concept—generally remain unaffected by the keyboard, since the volume of a note or sound is usually independent of the pitch or the "state" of the key press. While it is possible to influence the VCAs indirectly from the keyboard via envelope generators, the envelopes always follow the same contour, regardless of whether the key is pressed gently or struck forcefully.

To put an end to this "shortcoming" of conventional synthesizer systems, the touch controller for voltage-controlled music synthesizers was developed.

A synthesizer based on the "modular principle" is a musical instrument of "unlimited possibilities." Any "connection" is possible; nothing is forbidden. Therefore, it is obvious that the "Touch Controller" can, of course, also be used, deviating from its original purpose, to control any synthesizer elements, thereby revealing some musically appealing effects.

The circuit principle of the keystroke control is shown in Figure 1. In the resting state, capacitor C3 is positively charged via the changeover contact of the respective key. As soon as the key in question is pressed, the changeover switch opens momentarily—meaning no voltage is applied during this interval. The duration depends on the force (or, more accurately, the speed) of the keystroke.

During this time, C3 is discharged via R8, depending on the setting of P1 ("Touch Range"), more or less quickly. However, the voltage at C3 cannot fall below a value of +0.8V to +1.0V due to the voltage source. When the changeover contact reaches the "working" rail when a button is pressed, the instantaneous voltage at C3 is passed to the analog memory (sample and hold).

¹ See NPD (elsewhere in this chapter)

When a pressed button is released, the changeover contact is pushed back against the “idle” rail. As a result, C3 quickly charges to the value adjustable with P2 (Figure 2).

The design of the analog memory is similar to that of the sample-and-hold circuit in the FORMANT interface. To keep voltage drop (“drift”) within reasonable limits, this circuit also employs a “pre-” and “main” sample-and-hold stage. For this specific application, however, the drift problem is of only minor importance—unlike the sample-and-hold circuit for the KOV², where the usability of the entire interface depends on the stability of the sample-and-hold stage.

As in the FORMANT interface's sample-and-hold circuit, the FET switch (T2) is triggered by the gate circuit. In the quiescent state, a negative voltage of approximately -12 V to -14 V is applied to D1, almost completely blocking T2's drain-source channel (resulting in a resistance of several megaohms). Once a key is pressed, and following a delay adjustable via P7 (on the interface board), a positive voltage (+12 V to +14 V) is applied to D1 for the duration of the keypress; this voltage causes D1 to cut off. T2's drain-source resistance then drops to a few hundred ohms—meaning the switch is closed—and the (velocity-dependent) voltage present at the “Vor” sample-and-hold input is passed to C2. A simple op-amp voltage follower follows the analog storage stage, serving as a buffer and impedance converter.

Circuit Layout

The proposed PCB design (Figure 3) includes the 1.5 V voltage source, the analog memory, and the buffer stage. The board is dimensioned so that it can be mounted to the FORMANT interface board using spacers (similar to the triangle/sine converter for the LFOs—see Chapter 2). R8, C3, and D2 are mounted directly on the contact block, corresponding to the number of keys. The wiring is shown in Figure 4. The use of “solder strips” or “keyboard strips”—offered by some companies—simplifies wiring and improves the instrument's “portability.”

FETs T1 and T3 (use only “A” types) must undergo the FET test familiar from the FORMANT series before being installed in the circuit. Figure 5 shows the test setup again. The values for the corresponding source resistors can be found in Table 1. A “B” or “C” type may also be used for T2. The use of Bi-FET op-amps (see appendix on FET op-amps for FORMANT) was omitted for the sake of simplicity, especially since “extreme precision” (regarding drift and tracking) is not the primary priority for this circuit.

Calibration

(Complete interface required!)

First, check the operation of the analog storage circuit:

1. Disconnect connection Z. Connect the input of the analog storage circuit to ground and A (interface board) via a changeover switch; initially, set the switch to ground and press

² See also Digital Keyboard Controller (elsewhere in this chapter)

any key. The source voltage at T3 must be less than 4 V and must not change after the key is released.

2. Now switch the changeover switch to A; the source voltage at T3 must not change during this step. Press a key again. The source voltage must now be approximately 3.5 V to 4.5 V higher than before. The same measurement results must also be observed at the output of the buffer op-amp (IC1). This confirms the proper functioning of the sample-and-hold circuit.

To adjust the control voltage range, first reconnect Z. Turn the wiper of P1 towards ground. Then, press a key as quickly as possible while adjusting P2 so that the voltage at the buffer output does not exceed 5 V to 6 V.

Applications

The control voltage available at the output—ideally routed to a dedicated 3.5 mm jack socket on the rear or side panel of the keyboard housing—can be used primarily to drive the AM input of the FORMANT VCAs, allowing the keyboard to directly control volume³. This basic setup enables the creation of touch-sensitive, piano-like effects. A convincing simulation of the dynamic response of acoustic pianos can be achieved by additionally modulating the VCA with a "mixed" AD/AR envelope, a type typical of this class of instruments.

Furthermore, the touch-sensitive control voltage can be used for filter modulation (TM), frequency modulation (FM), or to vary the modulation frequency of a VC-LFO. Controlling VCFs is particularly interesting from a musical standpoint: "modulating" the cutoff frequency creates a rapidly changing timbre (depending on how the keys are struck), resulting in touch-sensitive "wah-wah" effects.

Tables

Table 1. The table allows for the easy determination of the corresponding source resistors (R_s) based on the measured voltage (U_s).

U_s (V)	R_s (k Ω)
0.20	22
0.25	18
0.3 ... 0.4	15
0.4 ... 0.5	12
0.5 ... 0.8	10

³ To avoid clipping, the "Gain" potentiometer of the DUAL VCA must be turned down accordingly.

U_s (V)	R_s (k Ω)
0.8 ... 1.2	8.2
1.2 ... 1.6	6.8
1.6 ... 2.0	4.7

Table 2. Technical specifications.

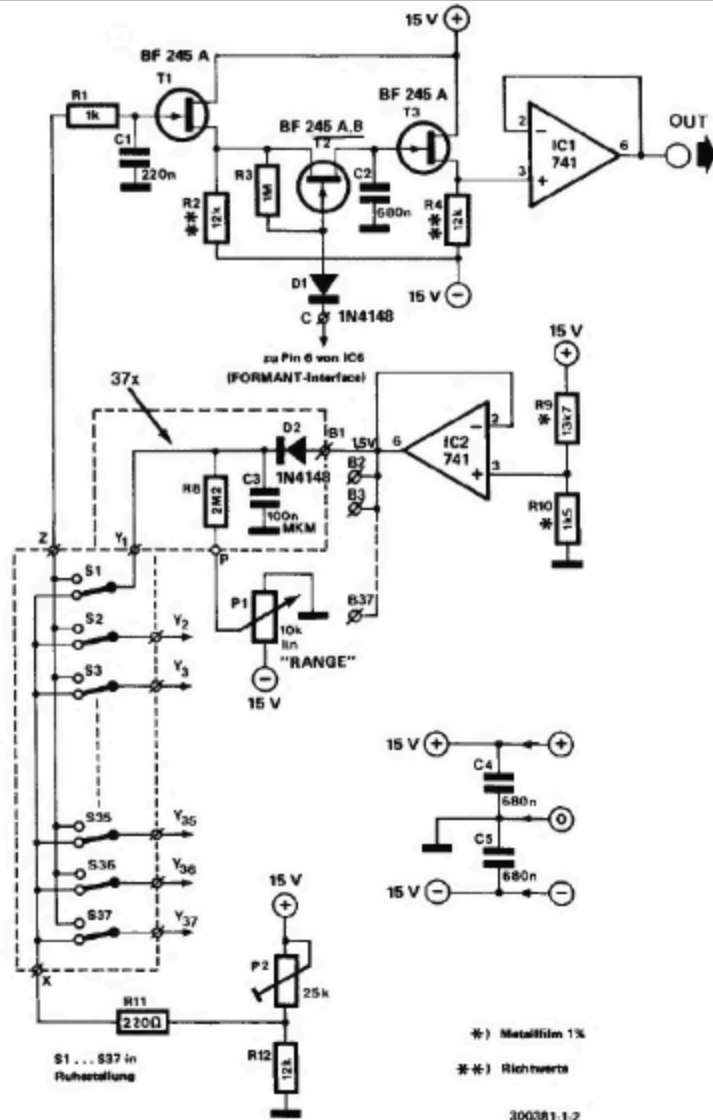
Input resistance at the analog memory:	approx. 10^{12} ohms
Drift (voltage drop, measured at the output of the analog memory):	typ. 0.15 mV/s
Max. input voltage at the memory:	+15 V
Control voltage range:	+0.8 V ... approx. +5 V (adjustable)
Characteristic:	approx. 12 dB/octave (when modulating an exponential VCA)

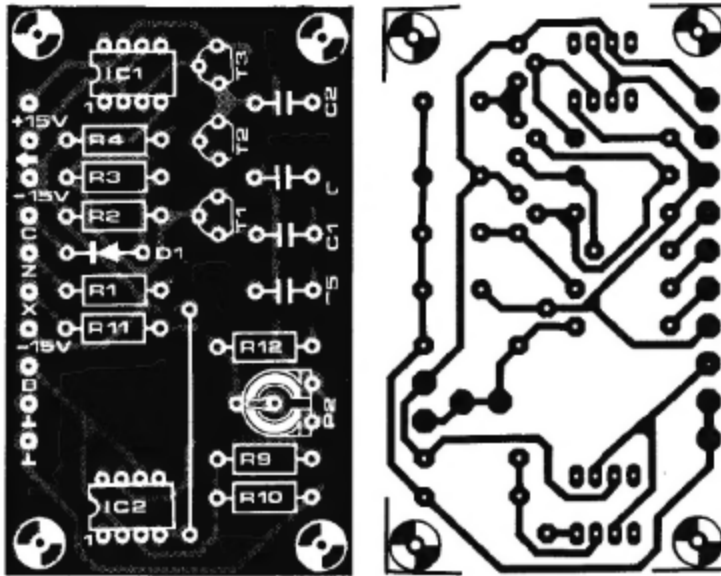
Figures

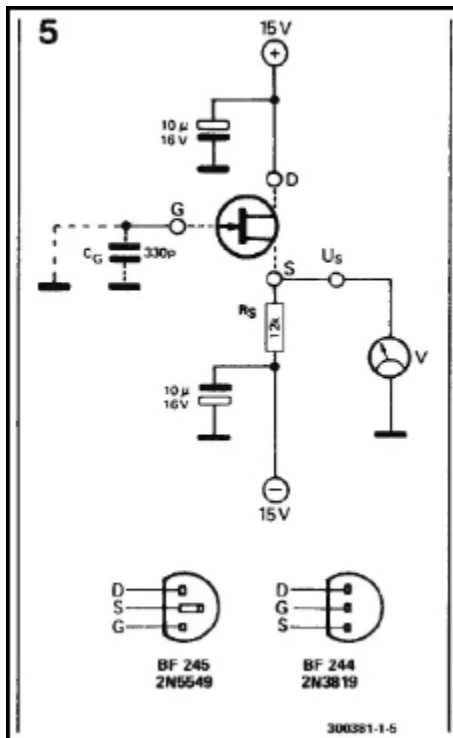
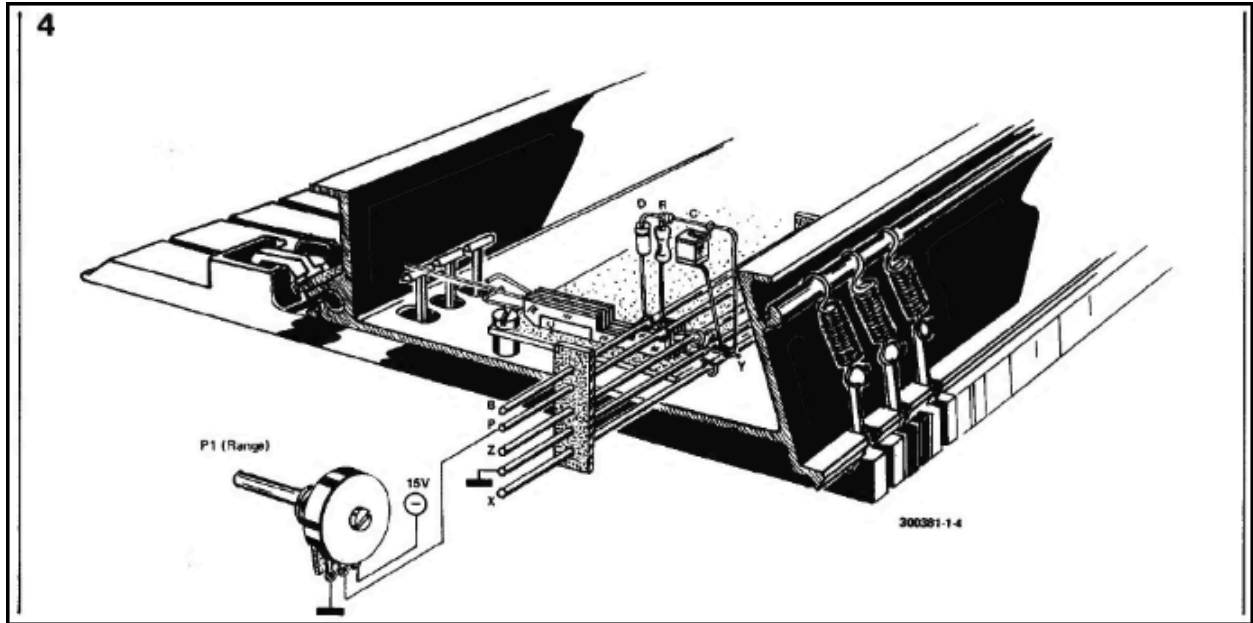
- Figure 1. Schematic diagram of the attack control circuit.
- Figure 2. Complete circuit diagram of the attack control circuit. To ensure proper operation, the marked resistors must be metal-film types. Source resistors R2 and R4, as well as FETs T1 and T3, must be carefully selected and matched. The test circuit described on page 22 of FORMANT Book 1 is suitable for this purpose.
- Figure 3. PCB layout and component placement plan for the circuit shown in Figure 2. The spacing of the PCB mounting holes matches that of the interface board, allowing both boards to be mounted together using spacers.
- Figure 4. Assembly drawing for the key contact wiring. For the sake of clarity, only the contacts and the extension for the keying control are shown.
- Figure 5. Test circuit required to determine the respective source resistors. For a detailed description of its operation, please refer to the FORMANT Book 1, page 22.
- Figure 6. Suggested front panel label for supplementing the FORMANT interface front panel.

**GATE-
SCHALTUNG
VOM
INTERFACE**









Parts list for Figure 2

Resistors (carbon film, 5%):

R1 = 1 k Ω

R2, R4 = 12 k Ω (approximate value; see text)

R3 = 1 M Ω

R8 = 2.2 M Ω (total of 37 units for a 37-key keyboard)
R11 = 220 Ω (see text)
R12 = 12 k Ω

Resistors (metal film, 1%):

R9 = 13.7 k Ω
R10 = 1.5 k Ω

Potentiometers:

P1 = 10 k Ω linear
P2 = 25 k Ω (trimmer)

Capacitors (low-leakage and low-loss):

C1 = 220n
C2, C4, C5 = 680n
C3 = 100n (total of 37 units for a 37-key keyboard)

Semiconductors:

T1, T3 = BF 245A (2N5549, BF 244A)
T2 = BF 245 A, B, C (2N3819, BF 244 A, B, C)
D1, D2 = 1N4148 (total of 37 units of D2 for a 37-key keyboard)
IC1, IC2 = μ A741C

Other:

2 spacer rollers
1 jack socket (3.5 mm) (see text)
Keyboard with independent changeover contacts

Sources for keyboards with independent changeover contacts (no guarantee):

BME Musikelektronik
H.P. Baumann
64, Rue de Welscheid
L-9018 Werken
Luxembourg

Keyboards:

No. Ky 003 3 oct. (36 keys)
No. Ky 105 5 oct. (60 keys)
No. Ky 023 6 oct. (72 keys)
2 changeover contacts and 1 make contact per key

England:
Phonosonics
22 High Street,
Sidcup, Kent, DA14 6EH, U.K.

Keyboards: Kimber Allen

Contacts: SKA
3-octave keyboard (37 notes)
4-octave keyboard (49 notes)
5-octave keyboard (61 notes)
Contact assemblies: type 4PS
3 pairs of contacts; each pair normally open – plus a single-pole changeover contact!

Further information is available upon sending 2 international reply coupons.

Additional Literature:

T. Orr, ETI String Thing,
Transcedent DPX, part 4,
Electronics Today International,
Vol. 8, No. 11 (Nov. 1979), p. 64 ff.